HIGH LEVEL DESIGN (HLD)

RAG-Based Healthcare Assistant using LangGraph & HITL

1. Overview

The project is a conversational healthcare assistant built using Retrieval-Augmented Generation (RAG), LangGraph, ChromaDB, and Groq LLM. The system retrieves healthcare-related information from PDF documents and generates grounded responses.

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2. Objective

- Reduce hallucinations in healthcare AI
- Generate accurate document-based responses
- Support conversational queries
- Implement Human-in-the-Loop (HITL) refinement

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3. System Architecture

User Query
↓
Conversation Memory
↓
Retriever
↓
Chroma Vector Database
↓
Relevant Chunks
↓
LLM (Groq)
↓
Generated Response
↓
Human Review (HITL)

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4. Major Components

Component	Purpose
PDF Knowledge Base	Stores healthcare documents
Chunking Module	Splits text into chunks
Embedding Model	Converts text into vectors

Component	Purpose
ChromaDB	Stores embeddings
Retriever	Retrieves relevant chunks
LangGraph	Controls workflow
LLM	Generates responses
HITL Module	Handles feedback refinement

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5. Technologies Used

- Python
- LangChain
- LangGraph
- ChromaDB
- HuggingFace Embeddings
- Groq API

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6. Key Features

- Conversational RAG
- HITL feedback loop
- Conversational memory
- Context-aware retrieval
- Healthcare response generation

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7. Future Enhancements

- Query rewriting
- Multi-document support
- Long-term memory
- Streamlit deployment